

Education

- Expected 2026 **Ph.D. in Mathematics**, Queen's University, Kingston, Ontario, Canada.
Advisor: Prof. Serdar Yüksel
- 2021 **M.Sc. in Mathematics**, Boğaziçi University, Istanbul, Turkey.
Advisor: Prof. Ümit Işlak
Thesis: *Edge Flipping in Graphs*
- 2019 **B.Sc. in Computer Engineering**, Boğaziçi University, Istanbul, Turkey.
Advisor: Prof. A. C. Cem Say
Thesis: *Modeling and Simulating Cellular Automata*

Research Interests

My research interests lie broadly in stochastic control, nonlinear filtering, and partially observed Markov decision processes (POMDPs). I am particularly interested in the robustness, stability, and approximation of control and filtering systems under partial information.

In discrete-time settings, I have studied the impact of model mismatch and quantization on the stability and convergence of belief-state dynamics, deriving explicit bounds on value functions and identifying conditions for near-optimality. In particular, I have investigated the Wasserstein and total variation continuity of filter kernels under model perturbations. I have also worked on robustness in the average cost setting and on unique ergodicity results for control-free models.

More recently, I have been focusing on continuous-time stochastic systems under partial observation, especially diffusion processes. I am studying robustness to prior mismatch and how discrete-time approximations via time and observation quantization can be used to analyze and control such systems.

Preprints

- Demirci, Y. E., Kara, A. D., Yüksel, S. Sensitivity of Filter Kernels and Robustness to Incorrect Transition and Measurement Kernel Perturbations in Partially Observable Stochastic Control. arXiv:2508.10658v3
- Demirci, Y. E., Kara, A. D., Yüksel, S. Refined Bounds on Near Optimality Finite Window Policies in POMDPs and Their Reinforcement Learning. arXiv:2409.04351

Publications

- 2025 Demirci, Y. E., Yüksel, S. (2025). Unique ergodicity of non-linear filters via reachability and uniform weak continuity. *Electronic Communications in Probability*, 30, 82, 1-15. <https://doi.org/10.1214/25-ECP732>
- 2024 Demirci, Y. E., Kara, A. D., Yüksel, S. (2024). Average Cost Optimality of Partially Observed MDPs: Contraction of Non-Linear Filters and Existence of Optimal Solutions and Approximations. *SIAM Journal on Control and Optimization*, 62, 6, 2859-2883. <https://doi.org/10.1137/24M1643736>
- 2023 Demirci, Y., Işlak, Ü., Özdemir, A. (2023). Mixing time bounds for edge flipping on regular graphs. *Journal of Applied Probability*, 1-16. <https://doi.org/10.1017/jpr.2023.3>
- 2023 Çiçeksis, A., Demirci, Y. E., Işlak, Ü. (2023). The Variance and the Asymptotic Distribution of the Length of Longest k -alternating Subsequences. *Discrete Mathematics and Theoretical Computer Science*, 25(1). <https://doi.org/10.46298/dmtcs.10296>

- 2023 Demirci, Y. E., Ekim, T., Yıldız, M. A. (2023). Defective Ramsey Numbers and Defective Cocolorings in Some Subclasses of Perfect Graphs. *Graphs and Combinatorics*, 39(1), 18. <https://doi.org/10.1007/s00373-023-02612-4>
- 2023 Demirci, Y. E., Işlak, Ü., Yıldız, M. A. (2023). On a cover time problem on a dynamic graph with steps at random times. *Statistics and Probability Letters*, 197, 109821. <https://doi.org/10.1016/j.spl.2023.109821>
- 2021 Demirci, Y. E., Ekim, T., Gimbel, J., Yıldız, M. A. (2021). Exact values of defective Ramsey numbers in graph classes. *Discrete Optimization*, 42, 100673. <https://doi.org/10.1016/j.disopt.2021.100673>

Conference Proceedings

- 2025 Demirci, Y. E., Kara, A. D., & Yüksel, S. (2025). Sensitivity of Filter Kernels and Robustness to Incorrect Transition and Measurement Kernel Perturbations in Partially Observable Stochastic Control. In *Proceedings of the 2025 IEEE 64th Conference on Decision and Control (CDC 2025), Rio de Janeiro, Brazil, December 10-12, 2025*
- 2025 Demirci, Y. E., Kara, A. D., & Yüksel, S. (2025). Wasserstein Regularity of Nonlinear Filters As Belief-MDPs, and Implications on Ergodicity, Optimality and Learning for POMDPs. In *Proceedings of the 2025 American Control Conference (ACC 2025), Denver, Colorado, USA, July 8-10, 2025*.
- 2024 Demirci, Y. E., Kara, A. D., & Yüksel, S. (2024). Average Cost Optimality of Partially Observed MDPs: Contraction of Non-Linear Filters and Existence of Optimal Solutions and Approximations. In *Proceedings of the 2024 American Control Conference (ACC 2024), Toronto, Canada, July 8-12, 2024*.

Honors and Awards

- 2019–2021 Graduate Fellowship, The Scientific and Technological Research Council of Turkey (TUBITAK) International Mathematical Olympiads (IMO) — Silver Medal (2012), Silver Medal (2011)
National Mathematical Olympiads of Turkey — Gold Medal (2012, 2010), Silver Medal (2011)

Scholarships

- 2023–2024 **Ontario Graduate Scholarship (OGS)**, *Queen's University*.
Value: \$15,000
- 2024–2025 **Ontario Graduate Scholarship (OGS)**, *Queen's University*.
Value: \$15,000
- 2025–2026 **Queen Elizabeth II Scholarship Science Technology**, *Queen's University*.
Value: \$15,000

Teaching Experience

- 2026 **Teaching Fellow**, *Queen's University*, Kingston, Ontario, Canada.
Course: Engineering Math Project
- 2025 **Teaching Fellow**, *Queen's University*, Kingston, Ontario, Canada.
Course: Introduction to Operations Research
- 2024 **Teaching Fellow**, *Queen's University*, Kingston, Ontario, Canada.
Course: Introduction to Linear Algebra
- 2023 **Teaching Assistant**, *Queen's University*, Kingston, Ontario, Canada.
Courses: Differential and Integral Calculus, Complex Analysis, Graph Theory, Control of Stochastic Systems
- 2022 **Teaching Assistant**, *Queen's University*, Kingston, Ontario, Canada.
Courses: Calculus I, Vector Calculus, Advanced Calculus, Linear Algebra, Introduction to Linear Algebra

- 2021 **Teaching Assistant**, *Queen's University*, Vector Calculus, Advanced Calculus.
- 2021 **Teaching Assistant**, *Bogazici University*, Differential Equations, Introduction to Probability and Statistics.

Talks

- November 2025 **Stability, Approximation, and Robustness of Optimal Policies in POMDPs**, *ISS Informal Systems Seminar*, McGill University, Montréal, QC, Canada.
- September 2025 **Unique Ergodicity of Non-Linear Filters via Reachability and Uniform Weak Continuity**, *Weekly Seminar on Stochastic Control*, Queen's University, Kingston, ON, Canada.
- July 2025 **Wasserstein Regularity of Nonlinear Filters As Belief-MDPs, and Implications on Ergodicity, Optimality and Learning for POMDPs**, *ACC 2025*, Denver, Colorado, USA.
- May 2025 **Stability, Ergodicity, and Optimal Control in Partially Observed Markov Processes: Wasserstein Contraction Approach**, *10th Meeting on System and Control Theory*, HEC Montréal, Montréal, QC, Canada.
- May 2025 **Wasserstein Perturbation Bounds on Filter Kernels in State Transition and Measurement Kernel Uncertainty**, *Weekly Seminar on Stochastic Control*, Queen's University, Kingston, ON, Canada.
- March 2025 **Robustness to Incorrect Models in Partially Observable Markov Decision Processes via Wasserstein Regularity of Filter Dynamics**, *Weekly Seminar on Stochastic Control*, Queen's University, Kingston, ON, Canada.
- October 2024 **Application of the Birkhoff-Hopf Contraction Theorem: Stability of Nonlinear Filters**, *Weekly Seminar on Stochastic Control*, Queen's University, Kingston, ON, Canada.
- September 2024 **Refined bounds on near optimality of finite window policies in POMDPs**, *Weekly Seminar on Stochastic Control*, Queen's University, Kingston, ON, Canada.
- July 2024 **Average Cost Optimality of Partially Observed MDPs: Contraction of Non-Linear Filters and Existence of Optimal Solutions and Approximations**, *ACC 2024*, Toronto, ON, Canada.
- February 2024 **Average Cost Optimality in Partially Observable Markov Decision Processes: Implications for Robustness and Optimal Policy Approximation**, *Weekly Seminar on Stochastic Control*, Queen's University, Kingston, ON, Canada.
- December 2023 **On Regularity and Ergodicity of Partially Observable Markov Decision Processes**, *CMS Winter Meeting*, Stochastic Control Theory and Applications, Montréal, QC, Canada.
- May 2023 **Unique Ergodicity of Non-Linear Filters**, *9th Meeting on System and Control Theory*, University of Waterloo, Waterloo, ON, Canada.
- March 2023 **On Unique Ergodicity of Non-Linear Filters**, *Weekly Seminar on Stochastic Control*, Queen's University, Kingston, ON, Canada.
- November 2022 **Average Cost Dynamic Programming Equations for POMDP**, *Weekly Seminar on Stochastic Control*, Queen's University, Kingston, ON, Canada.
- November 2020 **Edge Flipping in Graphs**, *Student Probability Seminar Series*, Boğaziçi University, İstanbul, Türkiye.
- July 2020 **Generating Random Graphs**, *Discrete Mathematics Days*, Boğaziçi University, İstanbul, Türkiye.
- March 2020 **Alternating Subsequences**, *Student Probability Seminar Series*, Boğaziçi University, İstanbul, Türkiye.
- November 2019 **Defective Ramsey Numbers in Graph Classes**, *Graph Theory Workshop IX*, Istanbul Center for Mathematical Science, İstanbul, Türkiye.